

Multiple Choice (10 marks)

1. What proportion of vegans is likely to be deficient in vitamin B 12?
 - (a) 3%
 - (b) 32%
 - (c) 66%
 - (d) 15%
2. Nitrites are added to processed meats mainly to prevent the growth of the following genus of microorganisms
 - (a) salmonella
 - (b) clostridium
 - (c) listeria
 - (d) campylobacter
3. Colonic microbiome:
 - (a) Is present in babies in utero
 - (b) Is a cause of eosinophilic oesophagitis
 - (c) Is easily changed by probiotic supplementation
 - (d) May be changed with consumption of oligo-saccharides
4. Which of the following diet-derived compounds have epigenetic effects?
 - (a) Butyrate
 - (b) Curcumin
 - (c) Genistein
 - (d) All of the options given are correct
5. What nutritional strategies will not improve performance during prolonged endurance type exercise?
 - (a) Carbohydrate loading
 - (b) Tapering
 - (c) The classic glycogen loading regime
 - (d) Exogenous carbohydrate ingestion during exercise

True / False (10 marks)

Answer True or False

6. True or False: Rickets of prematurity can be linked to hypocalcaemic convulsions, frusemide use, and maternal vitamin D deficiency.
7. True or False: A deficiency of folate can lead to megaloblastic anaemia, a condition characterized by the production of large, immature red blood cells.
8. True or False: Randomised controlled trials are the best study design to evaluate the effects of sugar on dental health.
9. True or False: Energy balance in a healthy person is typically achieved on a day-to-day basis without considering longer time frames.
10. True or False: During pregnancy, a woman should consume an additional 200 kCal only in the third trimester.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how infections or inflammatory states can affect biochemical indices of micronutrient status and explain why these changes might lead to misleading results.
12. Discuss the concept of goitrogens and analyze how certain foods, particularly those in the Brassica family, may impact thyroid health.

End of Paper